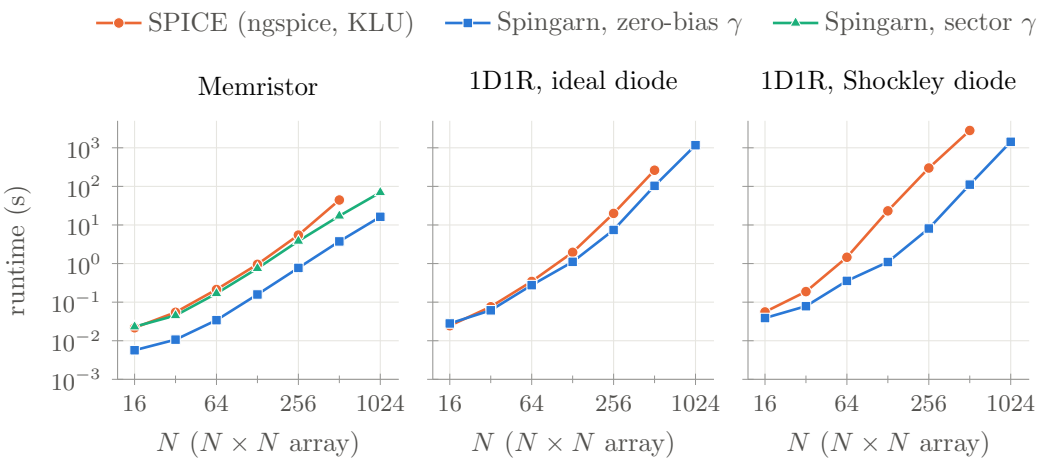


Figure 1: Runtime against array size for SPICE (ngspice-47, KLU) and Spingarn with fixed port resistances ($\alpha = 1$ on the memristor, 0.8 on 1D1R; CHOLMOD with AMD ordering), median over three random circuits. Zero-bias γ: $\gamma = R_{\min}$ per memristor; sector γ: $\sqrt{R_{\min}R_{\max}}$. On 1D1R the diode is on its own branch with port resistance $\sqrt{R_{\min}R_{\max}}$. SPICE writes the ideal diode as uramp$(V)/1\,\mathrm{m}\Omega$ and the Shockley diode ($I_s = 10^{-9}$ A, $n = 1$) as a behavioural exp source; on that case every SPICE run needed gmin stepping, and its 512×512 point is a single run. SPICE was not run at 1024×1024, where ngspice's KLU fails.

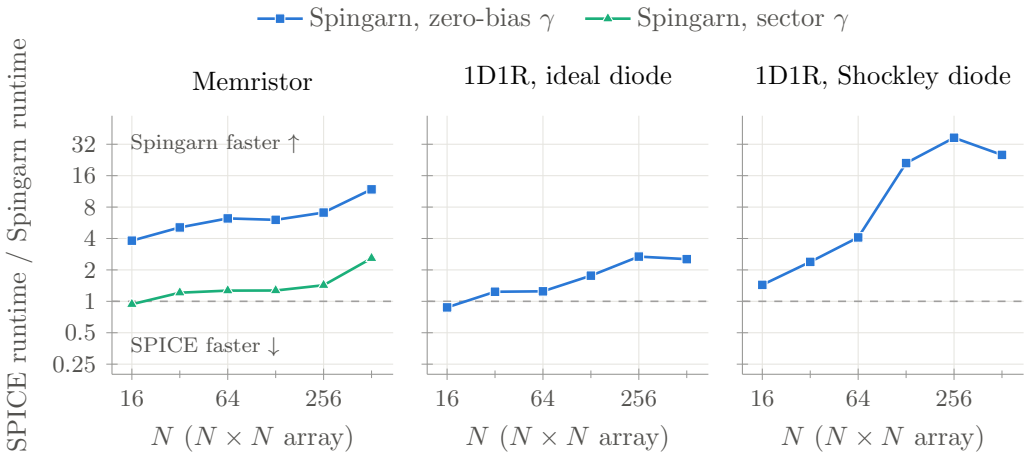


Figure 2: SPICE's runtime divided by Spingarn's, from the medians of the runtime figure.

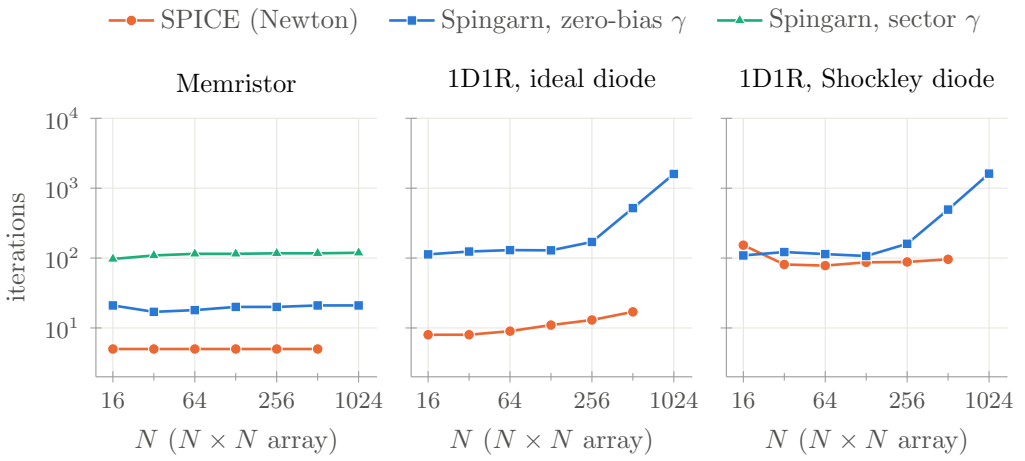


Figure 3: Iterations: Spingarn's, and SPICE's Newton iterations including fallback stages (gmin stepping).

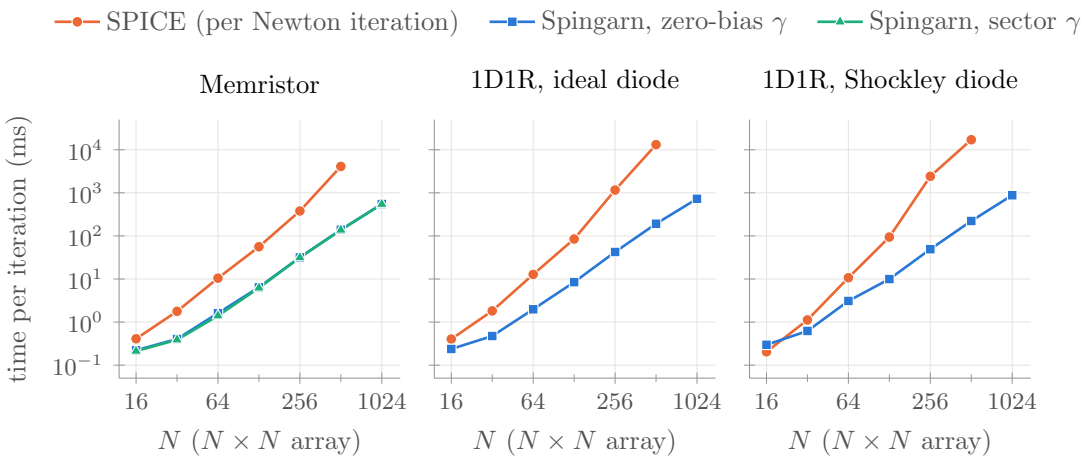


Figure 4: Time per iteration, one-time costs (netlist load and KLU reordering; Spingarn's assembly and factorization) excluded. On the memristor the two Spingarn variants coincide: they factorize the same matrix and do the same work per iteration.

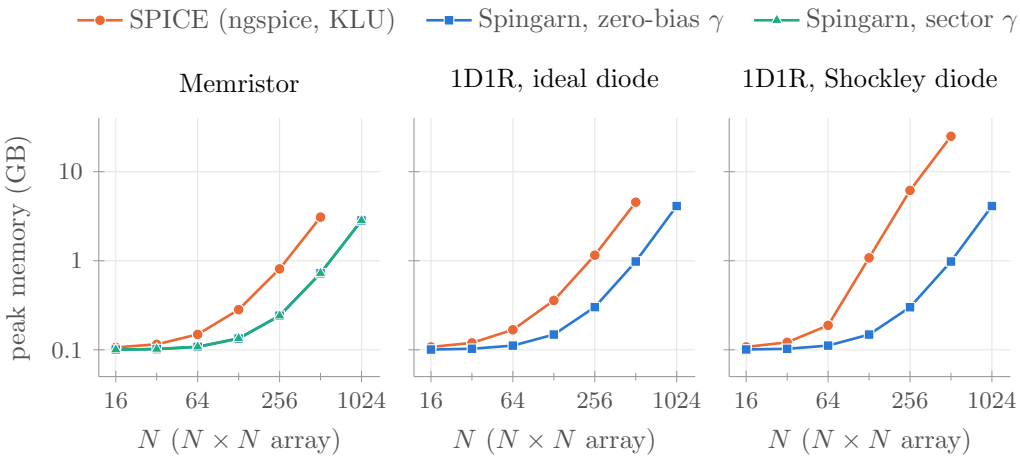


Figure 5: Peak memory of the process each run executed in. On the memristor the two Spingarn variants coincide.